

Auteur: Anna Voronovska
Studierichting: Informatica
Oefeningenreeks 4A nr. 3.12

$$\begin{aligned}\int 4x\sqrt[3]{3x}dx &= \int 4x \cdot (3x)^{\frac{1}{3}}dx = 4 \cdot \sqrt[3]{3} \int x^{\frac{4}{3}}dx = 4 \cdot \sqrt[3]{3} \cdot \frac{x^{\frac{7}{3}}}{\frac{7}{3}} + c \\ &= \frac{4 \cdot 3^{\frac{1}{3}} \cdot 3 \cdot x^{\frac{7}{3}}}{7} + c = \frac{4 \cdot 3^{\frac{4}{3}} \cdot x^{\frac{7}{3}}}{7} + c = \frac{4 \cdot \sqrt[3]{3^4} \cdot \sqrt[3]{x^7}}{7} + c\end{aligned}$$

References